令和 2 年度本試

1 小問集合

(1) 解答　ア $= 5$, イ $= 3$, ウ $= 6$

$$\begin{aligned}\frac{1}{\sqrt{3}} \div (-\frac{1}{2})^2 - \sqrt{6} \times \frac{\sqrt{2}}{4} &= \frac{1}{\sqrt{3}} \times 4 - \frac{2\sqrt{3}}{4} \\ &= \frac{4}{\sqrt{3}} - \frac{\sqrt{3}}{2} \\ &= \frac{4 \times 2 - \sqrt{3} \times \sqrt{3}}{2\sqrt{3}} \\ &= \frac{8-3}{2\sqrt{3}} \\ &= \frac{5\sqrt{3}}{6}\end{aligned}$$

(2) 解答　エ $= 1$, オ $= 2$

$$\begin{aligned}x^2 + ax - 6 &= 0 \\ (x+3)(x+B) &= 0 \\ (x+3)(x+B) &= x^2 + (3+B)x + 3B \\ x^2 + ax - 6 &= x^2 + (3+B)x + 3B\end{aligned}$$

$$\begin{aligned}a &= B + 3 \quad (1) \\ -6 &= 3B \quad B = -2 \quad (2)\end{aligned}$$

(2)を(1)に代入

$$\begin{aligned}a &= -2 + 3 = 1 \\ a &= 1\end{aligned}$$

$a = 1$を与式に代入

$$\begin{aligned}x^2 + x - 6 &= 0 \\ (x+3)(x-2) &= 0 \\ x &= -3, 2\end{aligned}$$

よってもう 1 つの解は2

(3) 解答　カキ $= -1$

$y = -\dfrac{1}{4}x^2$に$x = -3$を代入

$$y = -\frac{1}{4} \times (-3)^2 = -\frac{9}{4}$$

$y = -\dfrac{1}{4}x^2$に$x = 7$を代入

$$y = -\frac{1}{4} \times 7^2 = -\frac{49}{4}$$

変化の割合は$\dfrac{-\frac{49}{4} - (-\frac{9}{4})}{7 - (-3)} = \dfrac{-10}{10} = -1$

(4) 解答　ク $= 2$, ケ $= 3$

$$\frac{1}{2} \times (4 + 2) \times (4a + 1) = 11$$
$$3(4a + 1) = 11$$
$$4a + 1 = \frac{11}{3}$$
$$4a = \frac{8}{3}$$
$$a = \frac{2}{3}$$

(5) 解答　コ $= 7$, サシ $= 15$

	1	2	3	4	5	6
1	2	③	4	⑤	6	⑦
2	③	4	⑤	6	⑦	8
3	4	⑤	6	⑦	8	9
4	⑤	6	⑦	8	9	10
5	6	⑦	8	9	10	⑪
6	⑦	8	9	10	⑪	12

表より $\dfrac{14}{30} = \dfrac{7}{15}$

(6) 解答　ス $=$ ⓐ

平均値$x = \dfrac{1}{40}(2 \times 1 + 3 \times 2 + 4 \times 3 + 5 \times 4 + 6 \times 4 + 7 \times 6 + 8 \times 8 + 9 \times 10 + 10 \times 2) = 7$

中央値$y = 7.5$

最頻値$z = 9$

よって$x < y < z$

(7) 解答　セ $= 1$, ソ $= 9$

$$\angle \mathrm{ADG} : \angle \mathrm{AGE} = 1 : 3$$
$$\angle \mathrm{AGE} : \angle \mathrm{AFC} = 1 : 4 \Rightarrow \angle \mathrm{AGE} : \text{四角形}EGFC = 1 : 3$$

よって$\angle$ADG : 四角形$EGFC = 1 : 9$

(8) 解答　タチ $= 13$

高さを h、底面の対角線の交点を m とする

$$\frac{1}{3} \times 6 \times 8 \times h = 192$$
$$h = \frac{3 \times 192}{48} = 12$$
$$AM = \frac{1}{2}\sqrt{6^2 + 8^2} = 5$$
$$AO = \sqrt{12^2 + 5^2} = 13$$

2　連立方程式

(1) 解答　ア $= 2$, イウ $= 15$, エ $= 2$, オ $= 1$, カ $= 5$, キ $= 2$,

A さん

$y = ax + b$ に (7.5,1),(15,0) を代入

$$1 = 7.5a + b \quad (1)$$
$$0 = 15a + b \quad (2)$$

(1) $\times 2 -$ (2)

$$2 = 15a + b$$
$$- \quad 0 = 15a + b$$
$$2 = b$$

$b = 2$を(2)に代入

$$0 = 15a + 2$$
$$a = -\frac{2}{15}$$

よって$y = -\frac{2}{15}x + 2 \quad (A)$

B さん

$y = ax + b$ に (10,0),(15,1) を代入

$$0 = 10a + b \quad (1)$$
$$1 = 15a + b \quad (2)$$

$(2) - (1)$

$$\begin{array}{rl} & 1 = 15a + b \\ - & 0 = 10a + b \\ \hline & 1 = 5a \end{array}$$

$$a = \frac{1}{5}$$

$a = \frac{1}{5}$を(1)に代入

$$0 = 10 \times \frac{1}{5} + b$$
$$b = 2$$

よって$y = \frac{1}{5}x - 2 \quad (B)$

(2) 解答　$ク = 0, ケ = 4$

(A) と (B) を連立する

$$-\frac{2}{15}x + 2 = \frac{1}{5}x - 2$$
$$-2x + 30 = 3x - 30$$
$$5x = 60$$
$$x = 12$$

$x = 12$を (B) に代入

$$y = \frac{1}{5} \times 12 - 2 = 0.2 \times 12 - 2 = 2.4 - 2 = 0.4$$

(3) 解答　コサ $= 42$

A さん

$$y = \frac{2}{15}(x-30) = \frac{2}{15}x - 4$$

B さん

$$y = -\frac{1}{5}(x-40) = -\frac{1}{5}x - 8$$

A さんと B さんを連立

$$\frac{2}{15}x - 4 = -\frac{1}{5}x - 8$$
$$2x - 60 = -3x + 120$$
$$5x = 180$$
$$x = 36$$

A さんの式に代入

$$\frac{2}{15}(36-30) = \frac{4}{5}$$

$y = -\frac{2}{15}(x-c)$に (36,0.8) を代入

$$\frac{4}{5} = -\frac{2}{15}(36-c)$$
$$12 = -72 + 2c$$
$$2c = 84$$
$$c = 42$$

(4) 解答　シ $= 5$, ス $= 6$

$1 \times 4 + 0.8 \times 2 = 5.6$

3　場合の数

(1) 解答　ア $= 5$, イ $= 3$, ウ $= 3$, エオ $= 14$, カ $= 3$

$$6 + a + a - 4 = 12$$
$$2a = 10$$
$$a = 5$$

$$b = p + 2q + r + q + 2r + s = p + 3q + 3r + s$$
$$4 = -1 + 3c + 3 \times (-2) - 3$$
$$3c = 14$$
$$c = \frac{14}{3}$$

(2) 解答　キ $= 0$, ク $= 7$

e の隣を f とする

$$6 = 5 + 3e + 3f + 1$$
$$3e + 3f = 0$$
$$f = -e$$
$$d = e - e = 0$$
$$-6 \leq e \leq 6$$
$$5 + e \leq 6 \Rightarrow e \leq 1$$
$$-e + 1 \leq 6 \Rightarrow e \geq -5$$
$$\therefore -5 \leq e \leq 1 \Rightarrow \text{ 7 個}$$

(3) 解答　ケコ $= -3$, サ $= 0$

$$6 - (-2f - 3) = -7f - 18\text{(下から三段目一番左)}$$
$$9f = -27$$
$$f = -3$$

$$g = -f + 3 - 6$$
$$f + g = -e$$
$$-3 + g = -3g \qquad = 0$$

4 平面図形

(1) 解答　ア $= 8$, イ $= 5$

$CH = x$ とおくと

$$AH : AC = AC : AB$$
$$2x : \sqrt{5}x = \sqrt{5}x : 2$$
$$5x^2 = 4x$$
$$5x = 4$$
$$x = \frac{4}{5}$$
$$\therefore AH = 2x = \frac{8}{5}$$

(2) 解答　ウエ $= 45$, オ $= 8$, カ $= 7$

$$\angle ADB = 90^\circ \quad \angle ADE = 45^\circ$$

$$\begin{aligned} DB &= \sqrt{(AB)^2 - (AD)^2} \\ &= \sqrt{(AB)^- (AH)^2} \\ &= \sqrt{2^2 - (\frac{8}{5}^2)} \\ &= \sqrt{\frac{100}{25} - \frac{64}{25}} \\ &= \sqrt{\frac{36}{25}} \\ &= \frac{6}{5} \end{aligned}$$

$\triangle DAB$において

$$AE : BE = AD : BD = \frac{8}{5} : \frac{6}{5} = 4 : 3$$

$$\therefore AE = \frac{4}{7}AB = \frac{4}{7} \times 2 = \frac{8}{7}$$

(3) 解答 キ $= 5$, ク $= 2$, ケ $= 7$

$$OE = AE - AO = \frac{8}{7} - 1 = \frac{1}{7}, OF = 1 \text{より}$$

$$EF = \sqrt{(\frac{1}{7})^2 + 1^2} = \sqrt{\frac{1}{49} + 1} = \sqrt{\frac{50}{49}} = \frac{5\sqrt{2}}{7}$$